

What ever happened to meaning?

Michael Goodale and Salvador Mascarenhas
Institut Jean-Nicod
Ecole Normale Supérieure
Department of Cognitive Studies

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Author note: Forthcoming in *Behavioral and Brain Sciences* as a commentary on Futrell and Mahowald’s “How linguistics learned to stop worrying and embrace the language models.”

The target article contends that a convergence has occurred between linguistics and language models. Reading the piece, however, reveals that what is meant by “linguistics” is usage-based linguistics. Such a convergence should be surprising to no one: whichever strand of usage-based linguistics one has in mind, surely nothing could be more usage based than the linguistic capacities of state-of-the-art language models trained on massive corpora of language in use, with no language-specific biases.

A more interesting question is whether “generative linguistics” still has any relevance today. Futrell and Mahowald say yes, absolutely, if one is interested in studying programming languages. We submit instead that generative linguistics, despite its current lack of popularity in cognitive science, actually has a great deal to offer, and is brushed aside, as the authors do, very much to the detriment of the field. Here we focus exclusively on formal semantics, by which we mean the kind of mathematically sophisticated compositional semantics pursued in mainstream generative-linguistics departments. In short, what they’re doing over at MIT.

Despite the centrality of meaning to language, the authors only spare a few words on the topic: two paragraphs on the meanings of words and one on the principle of compositionality. The authors promote distributional semantics as the right theory of the meanings of content words, and directly contrast it with the mainstream in formal semantics, citing Heim and Kratzer’s (1998) textbook. But this contrast isn’t right. Formal semantics has never had much to say about the meanings of content words or the concepts that lie behind them. This is not an oversight, but an extremely wise methodological move by formal semantics. One couldn’t tell from reading the target article, but the question of what *concepts* are is an unmitigated quagmire. Reams of paper have been written on the many competing theories of concepts and word meanings: essential descriptions, sensory definitions, theory-theory, prototypes and stereotypes, conceptual roles, extensions. Evidently, the authors believe that distributional semantics and vector spaces somehow definitively settle all of these fraught debates, but they do not tell us how.

Compare with the approach from formal semantics. Word meanings are concepts. Nobody knows what on earth concepts are, but we’re certain that the concept LION needs to give us some sort of a procedure for telling whether something is or isn’t a lion. So we put a placeholder for that in our semantics, which *you* can fill in with your favorite theory of concepts. *We* will focus on combinatorial properties.

Which brings us to compositionality. The authors’ paragraph addresses no question of import. Instead, it focuses on the supposedly damning problem of “green tea.”

Here's what we take to be the authors' concern. "Green tea" doesn't just mean tea which is green, yet it isn't a completely opaque idiom. Therefore compositionality is gradable, and if this is true of most complex expressions, then language isn't all that compositional.

First, both "green tea" and full-on idioms like "kick the bucket" have perfectly compositional, literal interpretations, which can be quite useful, for example in gallows humor or to refer to tea which is green. Moreover, just because something *seems* gradable, it does not follow that the best way to think about *it* is in gradable terms. Truth can feel quite gradable, as manifested in expressions like "This is so true," or "This is far more true than that." Yet, the mainstream view in science as in philosophy is that truth is *not at all* gradable, but we have notions like *probability* and *verisimilitude*, which are in fact gradable and are connected to truth. Once we recognize that each phrase has a compositional interpretation *and* potentially a host of idiomatic uses, we can say: compositionality itself is non-gradable as the mathematicians propose, but for any expression which *also* has an idiomatic use, we can consider the *similarity* between the idiomatic meaning and the compositional meaning. We don't see how it matters what precise proportion of complex expressions *also* have idiomatic readings. And by the way, what proportion out of *what*, the infinity of possible complex expressions? At any rate: problem solved, as far as we can see, with just a little charity and common sense, and without breaking faith with mathematics.

There is in fact a wealth of work in the formal-semantics literature on this and far thornier cases. Even Montague (1970/1974), in one of the founding documents of the field, addresses some of these questions. For example, a "fake lawyer" isn't a lawyer who is fake (see Guerrini, 2024, for a recent treatment), nor is a "big flea" necessarily a "big creature."

Most importantly, there is a deep connection between how meanings are composed in natural language and how human thought works. A recent issue of this very journal demonstrated quite clearly that the language-of-thought hypothesis is a perfectly respectable and viable research program (Quilty-Dunn et al., 2023). The idea is to explain the nature of human thought by means of discrete symbolic representations which can be combined algebraically, something diametrically opposed to the approach recommended by Futrell and Mahowald.

Given that natural language does such a remarkable job of expressing the content of so many of our thoughts, it makes sense to wonder whether insights from formal natural-language semantics might bear on thought. Thankfully, there is a fair amount of work along these lines already, for example providing theories of modal representations (Phillips & Kratzer, 2024) or of reasoning and decision making (Koralus & Mascarenhas, 2013). This scholarship repurposes theories from formal semantics which are altogether inaccessible under a view that focuses exclusively on usage patterns and statistical distributions.

Futrell and Mahowald point out that "linguistics needs the rest of science." While this is certainly true, the rest of science also needs linguistics, and not just the very particular brand of usage-based construction grammar which the authors favor.

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